Non-LTE Line Formation: DETAIL and SURFACE - Oh no not again

1 Introduction

A few years ago, Jack Giddings and I provided a description of DETAIL and SURFACE (Butler and Giddings, 1984) in which the input was described in great detail (no pun intended) but which neglected to inform the reader how to compile the programs. Nor was there any description of the output. For completeness, these descriptions are included here and the formulae now available are to be found in the first appendix. The description of the GRIEM program has been omitted as it has outlived its usefulness. Thomas Schöning has made VCS tables available for almost all conceivable eventualities. Obtain them via my good self or the auspices of the CCP7 library.

Many of you will have noticed that the old description was almost but not completely accurate. Hopefully, there are less errors this time round but this is an uncertain world.

The programs have not evolved much in the intervening period. This will change shortly. However, since the changes are upwardly compatible it will simply be a good opportunity to present them in a later paper. To whet your appetitites the new versions will include a velocity field, ALI operators, some parallelisation for the Cray, be much faster (at the cost of a greater core requirement) and, in the case of SURFACE, provide a much more efficient rotational broadening option (the current one is such a hack I am almost embarassed).

2 Compilation

Well, you've obtained the sources from the CCP7 library (or even, Lord forbid from me) and you want to know what to do now. You should have three program files, BSFSTR.F, SURFACE.PRE and DETAIL.PRE. I've written .PRE to indicate that these are NOT FORTRAN programs, they must be preprocessed first. To do this compile BSFSTR.F (the only problem will be if your compiler objects to a CDC PROGRAM statement), the string processor. The program requires a data file containing the dimensions for the problem at hand. An example would look like this

```
DETA CRAY
 7000 2000   10  200  100  200 2500  600   12  13510000   90 5000   90 2300 8000
 8000
FREQSFREQLATOMSLLEVSXLEVSSLEVSINDCSINDRSINDFSINDLSDATASACTIVCROSFDEPTHCROSLXSECS
     MAXGRMXTRBMXALB
```

(BSFSTR does have defaults for all input values but these are totally useless, we deal with so many different models, I couldn't decide what a sensible default might be). The first three lines are the actual input, the last two are the strings they replace in the *.PRE sources. These are a bit more descriptive than they used to be and have the following meanings

FREQS	Total number of frequencies
FREQL	Number of linearized frequencies
ATOMS	Total number of atoms
LLEVS	Total number of L levels
XLEVS	Total number of X levels
SLEVS	Total number of S levels
INDCS	Number of collisions
INDRS	Number of radiative transitions (BF, BX and BB)
INDFS	Total number of free-free transitions
INDLS	Total number of linearized transitions
DATAS	Total number of data entries in data stack
ACTIV	Number of active L Levels
CROSF	Total number of cross section data
DEPTH	Obvious innit?
CROSL	Number of linearized cross sections (lines only)
XSECS	Number of bound free cross sections
	blank for historical reasons
MAXGR	Number of groups
MXTRB	Size of the microturbulence table
MXALB	Number of albedos

The PRE file must be attached to FORTRAN unit 12, while the actual FORTRAN source will be written to unit 13, the latter may then be compiled and run with an appropriate data set (many are available from the CCP7 library).

If you are building your own model it's always a good idea to use the verify option (see 3) as the last line of input will then be repeated giving an indication of how far the program has got. Dimension problems are amongst the easiest to find as the program states that a dimension has been exceeded and repeats the current value of that dimension. Change the dimension data set and repeat until all dimensions are large enough.

The actual dimensions used are given in the output following the words 'PHYSICAL SETUP' (see D). The output is repeated here together with the corresponding strings from the string preprocessor input. This is particularly useful for the 'CROSS DATA' which are printed BEFORE they are checked.

```
PHYSICAL SETUP
   RUN STATISTICS FULL PART
      ATOMS          2    2          ATOMS   ATOMS
      LEVELS        53   53          LLEVS   LLEVS
      LTE LEVELS     8               XLEVS
      Q.D. SERIES    3               SLEVS
      COLLISIONS   451               INDCS
      FREQUENCIES 2947 1134          FREQS   FREQL
      OPACITIES    325               INDRS
      RATES        325   98          INDRS   INDLS
      FREE FREE      6               INDFS
      CROSS DATA  1522 1304  218     CROSF   CROSL   XSECS
      INPUT DATA  1300               DATAS
      DEPTHS        90               DEPTH
```

In fact, BSFSTR supports keyword driven input of the dimensions (thanks to Dirk Husfeld) but I haven't used this myself and it has not been maintained so it may not work properly. Anyone interested?

The main problem with preprocessing at present is that some loaders complain (as they should) when they encounter common blocks of differing lengths. The blocks are buffered in DETAIL/SURFACE to be of the same length. Unfortunately, in the current versions this is done using the preprocessor which can cause problems for large cases since the amount of buffering needed can exceed 999999, the largest string provided for. The new versions use PARAMETER statements and so this will soon disappear. The only solution at the moment is to increase the dimensions of the variables in the buffered common blocks to reduce the size of the dummy arrays, certainly not a rewarding enterprise.

The keywords on the first input line give the target program and machine. If you are not lucky enough to have a CRAY, VAX or HP or a machine compatible with one of these, don't despair it is straightforward to add another option. First find out what functions on your machine provide the DATE, TIME and elapsed SECONDS. Then change the routines XDATE, XTIME and CPTIME to reflect these. You also need to know if your machine has single or double precision. If double then search the PRE files for the CDC keywords and duplicate the lines you find (unless your compiler has something against PROGRAM statements), if single then do the same, looking for and duplicating VAX keywords. That's it for the sources. The CRAY option shows places where library matrix inversion routines may be inserted if this is at all relevant. Finally, a suitable machine keyword must be put into BSFSTR to allow the preprocessing to take place.

3 Input

Both SURFACE and DETAIL have the same overall block structure. The overlay and archive files have been removed from the present versions. Communication is now (mostly) via common blocks. In the very latest versions most of this has been shifted to the subroutine calls in the solution code enabling parallelisation to be performed more easily.

The blocks are as follows:–

READ — in which the data are read in and checked. There are various levels of print-out available as an aid to the second of these. All data are read in free-field format so that zeroes must be included explicitly. If all the data requirements are met the next block is executed.

SETUP — from the data given in the previous step, all necessary atomic physics parameters are calculated. At this stage there is a division between background rates that are held fixed during the linearization process and those that are not. The last comment does not apply to SURFACE of course. Again, it is possible to obtain a great deal of information about the atomic data via the print options. If no problems arise then SURFACE will proceed directly to the calculation of the radiation field as the third step. Because of the larger number of options in DETAIL the third step is controlled by

CODE — after an initial compulsory Λ-iteration which must be specified (an estimate of the rates is needed for the linearization), linearizations and Λ-iterations may be freely mixed. This and other options are determined by keywords that are read by CODE.

All the options are now described in in terms of the input to DETAIL. The input for SURFACE may be identical but more probably will have Voigt profiles and many more line frequency points in order that they may be delineated accurately.

An important point to note at the outset is that the input is restricted to columns 1 to 80 but the programs see the input as a single stream. This can lead to problems for the unwary since a character in column 80 is followed directly by the character in column 1 on the next line. Always set ':V2' for the verify option (see below) when working on a new data set.

Data are introduced into the program by the use of keywords. The order of appearance of some keywords (and their associated data) is mandatory and these are indicated by (**) in the following. Other data need not appear but, if they do, they must be positioned appropriately. They are marked with a single asterisk. Still other data may occur anywhere within a given section of the input. These are described at the end of the section in question. Two methods control the amount of data read by a keyword. Either the number of expected quantities is given or a list of input values is terminated by a 0. This will become clearer later but note that a 0 (alone) is therefore not a valid label and that care must be taken to give the correct number of values. To aid in this, the program has three verification levels which may be changed at any time. The control character ':' in the first column interrupts the flow of data to pass instructions to the reading routines. Thus the level of verification is set by

:Vn

where n is 0, 1 or 2. The printout obtained, increases from none for n=0 to all input quantities together

with some internal data for n=2. Comments may also be interspersed throughout the data by the use of

:T 'This is a comment'.

The first section gives a set of continuum frequency points in descending frequency order. Thus the first keyword must be

FREQ**

labf_1	ν_1
labf_2	ν_2
$\vdots$	$\vdots$
labf_n	ν_n
0	

All labels, both here and throughout the code, consist of one to six characters and the frequencies are in Hz. Frequencies less than 10^6 are assumed to be wavelengths (they must still appear in the correct order) and are converted to frequencies. The current version of the program uses trapezoidal quadrature for the frequency mesh so that no particular choice is necessary, save for numerical considerations. It is also advantageous to include frequencies at either side of ionization edges of interest.

The levels and transitions of the ions under consideration are read in next. Ions described in this way may be calculated explicitly or may appear only as contributions to the opacity. For the ions that are included in the statistical equilibrium equations, each level must have at least one transition that connects it to another level. If this is not the case, the rates matrix **A** becomes singular. The program has no checks for this so the user should take care that all levels are connected.

Each element is introduced by the keyword

ATOM** Labat Z At. Wt. Abund

and is followed by a label for the atom, the charge, Z, of the lowest ion considered, the atomic weight and the abundance by number density relative to the total number density. A negative value for the abundance implies that the number read is the logarithm (to base 10) of the actual abundance.

The levels of the ions of this element are read in next and may take three forms

L** Labl_1 g_1 ν_1 Parent

X* Labx g ν Parent

S* Labs g_0 g_1 Ryd μ_ℓ n_0 n_1 Parent

After each of **L**, **X** and **S**, there appears a list of the different levels of the ion each list being terminated by '0'. **L** and **X** differ only in that **X** level populations are held fixed in LTE relative to the parent level while **L** levels appear in the rate equations explicitly. The data for these are consequently similar. Each **L** and **X** level has a label, a statistical weight g, a frequency and a label detailing the parent level. The frequency is the ionization potential of the level in Hz and the parent label for **L** and **X** levels should be that of the ground state of the next ion so that the transition frequencies are calculated correctly. At each call to **L**, the ionic charge Z is increased by one and it is assumed that the following levels belong to the next ion. Hence, the **X** and **S** levels following a given **L** belong to the same ion.

The **S** level descriptors differ from the other two because they represent upper state sums that appear only in the number conservation equations. The data are statistical weight factors g_0, g_1 such that the weight of a level with principal quantum number n is $\mathrm{g}_1\mathrm{n}^2 + \mathrm{g}_0$. The g_1 term allows for sums over the angular momentum quantum number ℓ and spin s; a constant (Ryd) and quantum defect μ_ℓ so that the energy of the levels, relative to their parent is

$$-\frac{\mathrm{Ryd}}{(n-\mu_\ell)^2}$$

(N.B. Ryd is Z^2 times the Rydberg constant for the atom). The sum extends from $n=n_0$ to $n=n_1$ and the series converge to the 'parent' level of the ion. The Rydberg constant for an atom of atomic weight AW is given by

$$\frac{\text{Ryd} = 109737.312 \times c}{1 + 1./(1836. \times \text{AW}.}$$

The input of levels is terminated by a **K** level which should be the ground state of the most highly charged ion in the model. It has no associated **X** or **S** levels but the same data as for an **L** level is required. Only the statistical weight and label are used however, since no ion lies above it in the model.

The program checks that all 'parents' appear in the next ion levels as they ought and that labels are not repeated within an atom. If they are correct it then proceeds to read the data that define the transitions. Each transition is of a given type **TY** which has three attributes. The first is the type of transition, of which there are eight; collisional ionization between two **L** levels, CBF; collisional excitation between two **L** levels, CBB, between an **L** level and an **X** level, CBX, and between an **L** level and a series of S levels, CBS. Similarly, there are radiative transitions RBF, RBB, RBX and RFF. The last of these refers to free-free transitions. These types are followed by two integers. The first defines the formula to be used to calculate the cross section while the second specifies the number of data entries that the program is to expect. A transition type is therefore given by

TY** type nformula ndata

and this is followed by further data that depend on the type and formula. The program has some of the more standard formulae built into it and they are described in **??**. If the formula number does not match any of the built-in functions, it is assumed to be user defined and the program calls a user subroutine of the correct type. They are UBOUND and UTABLE for radiative bound-bound transitions, UPHOTO for radiative bound-free transitions and UFREE for free-free transitions. There are also UCOLBB and UCOLBF for bound-bound and bound-free collisional transitions. Although the user may define these as he wishes, the rates and profiles used in this work are described in the following sections, and are available for general use. The program is unable to check that the formula numbers are permitted, so that they should be inserted carefully.

The first examples of the various types is that of **RBB** which has the following form

RBB** lab_l lab_u f_{lu}

(Note that the type must be repeated once. Thereafter any number of bound bound transitions of this type may follow). The two labels must belong to levels of the same ion and the first value is assumed to be the oscillator strength of the transition. Other data may also be included if necessary, for a Voigt profile, say. The frequency mesh for the line is determined at this point. It may be defined either in terms of a nominal Doppler width or explicitly in Angstrom units. These are set by

DL n $\delta\lambda_1$ $\delta\lambda_2 \cdots \delta\lambda_n$ for lambda

DB n $\delta\lambda_1$ $\delta\lambda_2 \cdots \delta\lambda_n$ for Doppler

where there are n frequency points at wavelength $\delta\lambda$ relative to the centre of the line. For the Doppler case a line temperature may be given and is the temperature assumed when depth independent Doppler profiles are adopted. Otherwise it is used solely in the definition of the frequency mesh and is introduced by the keyword

TL temperature.

The current mesh, as given by the last call to **DL** or **DB**, is used for all transitions thereafter until the mesh is redefined. The frequency at line centre is usually obtained from the difference between the frequencies of the levels given under the **L** and **X** keywords but the observed wavelength may be specified by the use of

CL TY RBB nformula ndata.

If **CL** is quoted, the line wavelength (in Å) should appear immediately after the label of the upper level.

The **CBB** transitions also have a lower level and an upper level which are followed by ndata data values. As with all entries, a type **TY** is terminated by a 0. **RBX** and **CBX** transitions are defined in the same way as for the **RBB** and **CBB** transitions respectively. However, there are now three labels, in that the lower and upper level labels are preceded by the label of a parent. This parent is not necessarily the same as that given in the **X** data. It is used to specify the level to which the rate is added in the rate equation. The form is therefore

$$\begin{array}{lllll} \mathbf{CBX}^* & \text{parent}_1 & \text{lab}_l & \text{lab}_u & \text{data} \\ & & \text{lab}_l & \text{lab}_u & \text{data} \\ 0 & & & & \\ \mathbf{CBX}^* & \text{parent}_2 & \text{lab}_l & \text{lab}_u & \text{data} \end{array} \quad \textit{etc.}$$

Note that the same parent applies to all transitions until the **CBX** is terminated by a 0 at which point the parent can be altered or a new **TY** may appear.

A **CBS** transition has the same form as the **CBX** and **RBX** transitions except that the first datum must be a power, IP, that is used to scale the collisional cross section of a level of principal quantum number n_0 by a factor of

$$\left(\frac{n}{n_0}\right)^{IP}$$

where n_0 is defined by the **S** level. The cross section should be that of the n_0 level and a sum is taken from n_0 to n_1. There is no upper level for this case and the parent should agree with the one given in the **S** data. Hence,

CBS* parent lab_l IP data.

The **CBF** and **RBF** transitions also use the parent form and again, it is not necessary for the parent to agree with the one given by the **L** data (to allow for inner shell processes). Thus **CBF** transitions are defined

CBF* parent lab_l data.

However, the radiative rates need a frequency mesh for the calculation of the integrals over the radiation field. This is done in terms of the mesh given under **FREQ**

RBF* parent lab_l labf_i labf_j data.

The frequency labels must be in descending frequency order and if the transition is to be included in the rate matrices, then the second label should correspond to a frequency just greater than the ionization potential, so that the integral is performed accurately.

The final type, **RFF** has only the parent level specified as it is calculated over the entire frequency mesh

RFF* parent data.

At this point, further atoms may be defined in the same way, beginning with the keyword **ATOM**. If the atomic data have been completed, however, the next word must be

MULTI** n

and the first n atoms are included in the statistical equilibrium calculations. Next comes

LINEAR** n_1 n_2 $\cdots$ 0.

LINEAR is an instruction to include the radiative rates n_1, n_2, *etc.* as linearized transitions. The numbers n_i are given by the program when the verify option is used. Thus it is necessary to verify the data initially, not only to provide a check but also to obtain the numbers assigned to the radiative transitions.

The temperature-density structure must be input next. It is introduced by the keyword

MODEL**

and this section must contain a

STRUCTURE** ND.

ND is the number of depth points. There is then a title of up to 40 characters

TITLE** This is a title.

and ND values of:

an integer giving the current depth point;
the integrated mass to the current depth;
the temperature;
the total number density;
the electron density;
and two further numbers, which are not used and so are arbitrary (originally Auer/Mihalas atmospheres were used in which b factors were output in these columns)

In addition, a 40 character title for the current calculation may be set by

TITLE This is another title.

TEFF T_{eff} is obvious.

GRAVITY log g is the surface gravity.

If log g is greater than ten, the logarithm of the given value is taken.

The hydrogen and helium abundances may be defined by the use of

YHE Y_{He}

and the abundance of any elements by

ABUNDANCE labat_1 abund_1 labat_2 $\text{abund}_2 \cdots 0$.

As usual, negative values imply that the logarithm of the abundance has been given. Y_{He} is the helium abundance. The hydrogen abundance is calculated from this value as

$$1 - Y_{He}.$$

Abundances for any given element may be read in more than once and the last one in wins. This means that to set a new abundance while using populations from a previous run as starting values the command must appear *after* the populations have been read. This also applies to things such as microturbulence and albedos which appear in the populations file.

A microturbulent velocity in kms^{-1} may be defined by

TURBULENCE v_T.

Finally for this section, the populations of some, or all, of the levels may be read in. This provides a restart facility or allows the use of populations from other calculations. The keyword is

POPULATIONS ND_0

where ND_0 is the number of depth points for which populations are available. ND_0 must equal ND. Next comes another title

TITLE Yet another title.

and then an **ATOM** card

ATOM labat Z At. Wt. Abund

(see the earlier discussion). An **L** card gives the level for which the next ND_0 numbers are populations

lab_l g ν parent
pop_1 $\mathrm{pop}_2 \cdots \mathrm{pop}_{ND_0}$

lab_l g ν parent
etc.
0

A new atom may follow. If a level or an atom appears that is not included in the current calculation, the data for that level (atom) are read but ignored.

The structure may be followed by any of

OPACITY*

TABLE*

PRINT* p_1 p_2 p_3 p_4

FIX* labat lab_l lab_u

GO
There is a facility to input opacities other than by explicitly giving the atomic data. The user may add a subroutine UOPAC that defines the opacity and emissivity on the frequency mesh. The subroutine is called only if **OPACITY** is included. Similarly, the calculation of detailed Stark-broadened line profiles is expensive. Therefore, the subroutine UTABLE is available to allow profiles obtained from accurate line-broadening theories to be stored in tabulated form outside the program. UTABLE would then read in these profiles and interpolate from them onto the current temperature, density and frequency grids. **TABLE** ensures that UTABLE is called.

The print options are such that the populations (p_1), collision rates (p_2), foreground and background photoionization cross sections (p_3) and background opacities (p_4) are printed at the first and every p_ith depth point.

The keyword **FIX** states that all levels of atom labat which lie between the level with label lab_l and that with lab_u have their populations held at their initial values, even if they have been declared in the **MULTI** command. This has the effect of forcing two overlapping lines to be calculated at the same frequency points, even if one of the lines belongs to an atom which is to be held fixed. It also allows only part of an atom to be solved for explicitly.

If the word is **GO** the program calculates all the data – collision rates, opacities, cross sections – which are to be held fixed throughout the statistical equilibrium (SE) calculations. Any other keyword causes the program to terminate with an error message.
SURFACE then proceeds to generate the surface radiation field without more ado. For DETAIL, further keywords are read in CODE and are as follows.

The print levels throughout the SE calculations may be changed at any time, again by the keyword

PRINT p_1 p_2.

This has the same effect as before for p_2, which refers to the active radiative rates and the populations. The radiation field is printed if p_1 is greater than 0. The amount of information increases with p_1 to a maximum for $p_1 = 3$.

SCRIBE The populations at any stage of the computation may be output to fortran unit 7.

There are two different SE calculations. The first is a Λ-iteration and begins with the keyword

DRIVE.

At this stage, it is possible to hold the bound-bound transition rates fixed by specifying **NOLINE** for a single iteration. **NOLINE** must be respecified if the line rates are to be fixed for further Λ-iterations. **NOLINE** is disregarded by the linearization option. The second method is initiated by

RYBIKI n_{it} n_{sor} Ω TOL

where n_{it} is the number of linearizations, n_{sor} is the number of iterations in the approximate solution for the changes in the rates δZ_t, Ω is a parameter that controls the amount of successive over relaxations (sor) — 1.67 is generally found to be suitable — and TOL is a tolerance, which may be fairly coarse.

DRIVE must, of course, appear first, since the Rybicki linearization needs an estimate of the current values for the radiation field *etc.* However, thereafter, **DRIVE** and **RYBIKI** may appear in any desired order. It should be noted that the two systems are mutually inconsistent, unless all transitions are linearized. Hence, the Rybicki solution should prevail in the latter stages. The program provides plots at each overall iteration, so that the convergence of the solution may be seen clearly.

QUIT This keyword terminates program execution gracefully. Any other input causes a termination with an error message.

4 Output

A sample output is provided in D. It has been shortened somewhat to save space but shows all the essential aspects of a typical DETAIL output. On occasion timings of the various routines are given on the far right indicating when a particular section has finished its job. The times are elapsed CPU seconds.

The first section will be printed if a ":V2" appears at the start of the input data set. This should always be the case when developing a new model atom but can be switched off for production purposes. It is simply a repeat of the input data enabling these to be checked (we all make mistakes). The continuum frequency grid is followed by the model atoms with their L, X and S levels and then the transition information. For the latter, the data are repeated and the wavelength at line centre (BB) or at the ionization edge (BF) is also given. "MULTI" then says how many atoms are to be included in the statistical equilibrium equations. Next each linearized transition is listed coupled with the number of the transition in the input data set. There follow two further lists, the background and then the linearized transitions. The original transition number is given in the background list together with the levels involved and the wavelengths. An initial run with "LINEAR 0" saves the user from having to count the transitions by hand, speeding up the choice of transitions to be linearized. The numbering in the list of linearized transitions is important as later plots and tables in the RYBIKI section refer to the transition number in this list and not to the original order in the input data set which is what is needed for the LINEAR command. The number of the first and last frequencies for any given transition is also listed here which may help in finding the radiation field should it be needed.

The model atmosphere is printed out next including any populations which may have been read in and any comments added to the model atmosphere file. This is followed by the size of the problem described earlier, the print options are listed together with a brief description of the problem at hand. Of particular interest is the list of "FLOATING LEVELS". The numbering is that used by DRIVER for its plots and tables.

The sections that follow are switched on by the PRINT option. The LTE populations and Saha factors appear first for each requested depth. The Saha factors are normalized so that the K level of each atom has a factor of one which can be a help when trying to find information for a particular level. For each depth requested three tables then follow. The first contains the collisional rate coefficients C_{ij} in Mihalas (1978) for all transitions. Next comes information for transitions not included in the statistical equilibrium equations including all free-free transitions. The numbers in the introductory line are the depth index, the number of the transition in the appropriate list, the levels involved including their order in the level list and lastly the indices of the first and last frequencies. The cross sections are printed in this order which means that the cross section at the edge (smallest frequency) is generally the last number in the list.

These background opacities are summed and printed immediately since the populations are known (either they've been read in or are set to LTE values). For each frequency in the grid, two quantities are printed, call them C and E. C includes the electron scattering contribution and the total opacity and emissivity at this frequency are given by

$$\chi = (C - E)/\rho \tag{1}$$
$$\eta = 2h\nu^3/c^2 E/\rho \tag{2}$$

ρ being the density. Cross section information for active transitions then follows and the SETUP section closes with a repeat of the model atmosphere structure.

This output can be voluminous but is extremely useful for debugging new model atoms since obvious errors (negative or very large collision rates, cross sections) can be spotted immediately.

With "PRINT 3 1", the output from the DRIVER and RYBIKI subprograms is a maximum (and once more extremely voluminous). The minimum output is the plot from each iteration. For the DRIVER, the surface flux is printed for all frequencies. Frequency, wavelength, Eddington flux, the mass at which an optical depth of 1 is reached, total optical depth and value of the albedo are tabulated. The mass can be used to find the nearest depth point in the atmosphere from which other information, such as the rates or populations of interesting transitions, may be obtained. This is an aid in determining the dominant rates at the depth of line formation giving an idea of how a transition is formed. The fluxes are followed by a plot of the logarithmic relative changes in the populations as a function of depth. Each level is keyed by its index number in the plot. The changes should become smaller with increasing depth as LTE is then a good approximation. The surface is at the top for the DRIVER plot but at the bottom from the RYBIKI iteration allowing them to be distinguished at a glance.

The plot allows those levels to be picked out with unusually large corrections and the depths at which problems occur. This information, coupled with the optical depth information from the surface flux table can give clues as to the underlying physics of numerical problems. The most obvious is that of an ion being calculated at too low or too high a temperature leading to loss of significant figures in the solution of the equations of statistical equilibrium. Inconsistencies in the current model atom compared with that used in the model atmosphere can also lead to numerical instabilities caused by population inversions in the higher levels and subsequent emission. A good solution here is to start from LTE populations.

For each depth, all radiative rate coefficients and net rates followed by the populations are then tabulated. Mihalas writes the rate coefficients as R_{ij} while the net rates are defined by $R_{ij} - (n_j/n_i)R_{ji}$, thus a negative rate implies a net downward rate.

The surface flux for a RYBIKI iteration includes only the linearized frequencies but otherwise the same information as the DRIVER listing. With print option 3 line profiles are also given, flux versus wavelength. The fluxes are normalised simply to the maximum flux within the line. This maximum flux is printed as HMAX in the line header where the mass at the depth of formation (TAUL), maximum optical depth (TAUC) both at line centre and the frequency indices are given. For broad lines, the continuum can change significantly over the line profile which need not therefore be symmetric.

The rate and net rate coefficients come next. The numbering here corresponds to the order in the plot so that troublesome transitions may be found readily. Directly after the rates come the populations together with the b factors, referred to the *ground state of the next ion*. Other definitions are possible so the user should take care when conducting comparisons. As mentioned earlier, the plot which comes next has the innermost point of the atmosphere at the top where the changes should be small. Plotted are the logarithmic changes in the rates, a negative sign indicating that they have decreased during the iteration cycle. The maximum changes in the net rates rates which are tabulated are rarely useful. The depth index at which the maximum occurs is given together with the change for each linearized transition. I find the plot much more helpful.

The keyword "SCRIBE" if present enables the populations to be written to FORTRAN unit 7 which may be stored for use with SURFACE or as input to further DETAIL runs as necesssary. The format is straightforward, the interesting part being the populations which are output for each active level at all depths.

The SURFACE output is a subset of the DETAIL output and will not be described further. However, two files are output as two formal solutions are in fact performed, one with the other without line opacities. This enables Eddington fluxes to be written to FORTRAN unit 7 as with earlier versions while

normalised (continuum=1) fluxes are written to unit 70 (as a sort of mnemonic aid) thereby making the measurement of equivalent widths much easier.

A Formulae available in DETAIL for Atomic Data Specification

The following is a list of the different formulae that are currently available within DETAIL for the specification of atomic data. They are given in terms of the type number referred to in the previous section.

RBF

In these formulae, the variable

$$x = \frac{\nu_o}{\nu}$$

is the ratio of the frequency at the ionization edge, ν_o, to the frequency ν. The α_i are parameters where, in general, α_o is a measure of the threshold cross section (x=1).

1.

$$\alpha_o x^3$$

2.

$$\frac{\alpha_o}{x^p}$$

3.

$$\frac{\alpha_o}{x^p} + \frac{\alpha_1}{x^q}$$

4.

$$\alpha_o \left[\alpha_1 + (1 - \alpha_1)\, x\right] x^{\alpha_2}.$$

The previous formulae are generalizations of the hydrogenic form given next while type 4 is due to Seaton (1958).

5.

$$2.815 \times 10^{29} \frac{Z^2}{n^5} \frac{g_{II}}{\nu^3}$$

in which n is the principal quantum number and g_{II} is the bound-free gaunt factor (Mihalas 1978 p 99).

6.

$$\alpha_o e^{(\alpha_1 + \alpha_2 \nu)}$$

7.

$$e^{[\alpha_o + \ln(\nu)(\alpha_1 + \alpha_2 \ln(\nu))]}.$$

These last two allow the cross sections to be fitted by linear least squares fit procedures (see *section*4)

RBB

1.

$$\frac{1}{\sqrt{\pi}\Delta\nu_D} e^{-x^2}$$

where $\Delta\nu_D$ is the Doppler width in frequency units

$$\Delta\nu_D = \frac{\nu_o}{c} \left[\frac{2kT}{M} + \xi^2\right]^{\frac{1}{2}}$$

and k is Boltzmann's constant, M is the mass of the particle, ξ is the turbulent velocity. The variable x is the displacement of the frequency ν from the frequency at line centre ν_o in terms of the Doppler width

$$x = \frac{\nu - \nu_o}{\Delta\nu_D}.$$

Note that this is depth dependent as the temperature at the current depth, T, is used.

2.

$$\frac{1}{\sqrt{\pi}\Delta\nu_D} H\left(a,x\right)$$

where H(a,x) is the Voigt profile

$$H\left(a,x\right) = \frac{a}{\pi}\int_{-\infty}^{\infty} \frac{e^{-y^2}}{\left(v-y\right)^2 + a^2}\, dy$$

and a is the damping width which is parameterized as

$$\Delta\nu_D a = \Gamma_1 + N_e(\Gamma_2 T^p + \Gamma_3)$$

and hence can allow for pressure broadening. The usual definition for a differs from this by a factor of 4π. Therefore, all the Γ parameters should be divided by 4π before they are input to the program.

3. The Stark profiles of Griem (1960) may be fitted numerically to the following form up to some limiting electron density N_e. The Stark width $\Delta\nu_S$ is given by

$$\frac{1}{\Delta\nu_S} = \frac{c}{\nu_o^2}(b^2 - a^2)Z^5\Delta$$

in frequency units and

$$\Delta = 5.5 \times 10^{-15} a^4 b^4 \times 1.25 \times 10^{-9} N_e^{\frac{2}{3}}.$$

For

$$\Delta\nu_S < 0.194\Delta\nu_D$$

the depth dependent Doppler profile is used. Otherwise, with

$$x = \frac{\nu - \nu_o}{\Delta\nu_S},$$

the profile is

$$\frac{TH(x)}{\Delta\nu_S}$$

and $TH(x)$ is the fitted function.

4. The Lorentz width is

$$\Delta\nu_L = \frac{\Gamma}{4\pi}.$$

Hence if

$$x = \frac{\nu - \nu_o}{\Delta\nu_L}$$

the profile is

$$\frac{1}{\pi\Delta\nu_L}\frac{1}{1+x^2}.$$

5. The depth independent Doppler profile is of the same form as type 1 above except that the fiducial temperature T_{LINE}, mentioned in *section*1 is used rather than the 'physical' temperature.

RFF

There are only two built-in functions of this type. Both are based on the hydrogenic form, which is the first type

1.

$$3.69 \times 10^8 \frac{N_e}{T^{\frac{1}{2}}} Z^2 \frac{g_{III}(\nu, T, Z)}{\nu^3}$$

and g_{III} is the free-free gaunt factor.

2. The second form accounts for the opacity from an infinite series of LTE levels from a given value of the principal quantum number n, upwards. The frequency of the ionization potential of the n level is therefore,

$$\nu_n = 3.29 \times 10^{15} \frac{Z^2}{n^2}$$

(3.29×10^{15} is an approximate Rydberg constant in frequency units) and the smaller of the two frequencies, ν and ν_n, is chosen for the exponential factor, *i. e.* ν_n is the cutoff frequency

$$\nu_{min} = \min(\nu, \nu_n).$$

The opacity is then

$$3.69 \times 10^8 Z^2 \frac{N_e}{T^{\frac{1}{2}}} \frac{\left(e^{\frac{h\nu_{min}}{kT}} - 1\right)}{\nu^3}.$$

CBF

1. Seaton (1962) has given a formula for the collisional ionization cross section in terms of the photoionization cross section at threshold, α_o, which yields a rate

$$1.55 \times 10^{13} \alpha_o \overline{g} N_e \frac{e^{-u_o}}{u_o T^{\frac{1}{2}}}$$

where

$$u_o = \frac{h\nu_o}{kT}.$$

The $\overline{g}$ factor is Z dependent, being 0.1, 0.2 or 0.3 for Z=1, Z=2 and Z$\geq$3 respectively.

2. This and the next form assume simply that the cross section is inversely proportional to the energy or that it is constant. Upon integration over the Maxwellian distribution the rates are then found to be

$$5.436 \times 10^{-11} \frac{N_e}{T^{\frac{1}{2}}} e^{-u_o} \alpha_1$$

for the former case and

3.

$$5.436 \times 10^{-11} \frac{N_e}{T^{\frac{1}{2}}} e^{-u_o} \alpha_o \left(1 + u_o\right)$$

for a constant cross section.

CBB

The formulae for this type of transition follow the same pattern as for the CBF type. Van Regemorter (1962) gives a form for the collisional excitation cross section that is similar to the Seaton collisional ionization cross section. It is proportional to the oscillator strength, f_{ij}, since bound-bound transitions are involved. The rate is

1.

$$5.465 \times 10^{-11} \frac{N_e}{T^{\frac{1}{2}}} \left[14.5 f_{ij} \left(\frac{\nu_H}{\nu_o}\right)\right] e^{-u_o} \Gamma$$

if ν_H is the ionization potential of hydrogen. Γ is the maximum of

$$\Gamma = \max\left[\overline{g}, 0.276 e^{u_o} E_1(u_o)\right]$$

and $E_1(x)$ is the first exponential integral

$$E_n(x) = \int_1^\infty \frac{e^{-xt}}{t^n} dx.$$

The number $\overline{g}$ is found to be 0.2 if the principal quantum number changes during the transition and 0.7 if it does not.

2. The same two forms are provided here as in the CBF case giving the same rates

$$5.465 \times 10^{-11} \alpha_1 \frac{N_e}{T^{\frac{1}{2}}} e^{-u_o}$$

and

$$5.465 \times 10^{-11} \alpha_0 \frac{N_e}{T^{\frac{1}{2}}} e^{-u_o} (1 + u_o).$$

B Utility Routines Optionally Available to DETAIL and SURFACE Users

In the course of this work, many atomic data have been gathered; so that they might be accessed the routines UPHOTO, UCOLBF and UCOLBB were written. This section details the various formulae for ease of reference. It also contains a brief outline of two routines, UBOUND and UTABLE, that allow the Stark profiles for hydrogenic ions to be obtained according to the theory of Griem (1960). The type numbers of *section*1 are again referenced.

UPHOTO

11. Butler's (1984) photoionization data may be fitted to the quantum defect formulae reviewed by Seaton (1983). These are rather complicated generally, but at low energies the matrices that appear in the theory have only one element. The formulae become easy to manipulate in this region and are described here. However, it should be noted that a code which uses the full theory is available .

For a frequency ν the ejected electron has energy E,

$$E = EFE - IP \qquad \text{in Ryd}$$

the ionization IP should be in Rydberg units while the transition energy EFE may be calculated from

$$EFE = \frac{\nu}{cRyd}.$$

The rydberg for an atom of atomic weight AW is

$$Ryd = \frac{109737.312}{1 + \frac{1}{1836 \times AW}} \qquad \text{cm}^{-1}.$$

The fits have a reference energy E_{ref} so x is defined by

$$x = E - E_{ref}$$

and used to calculate

$$R = \sum_{i=1}^{IPP} a_i x^{i-1} + \sum_{j=1}^{NPOLE} \frac{b_j}{E - E_j} \tag{3}$$

$$G = \sum_{i=1}^{IPP} c_i x^{i-1} + \sum_{j=1}^{NPOLE} \frac{d_j}{E - E_j} \tag{4}$$

$$\tag{5}$$

with E_j, a_i, b_j, c_i, d_j as parameters. The cross section σ is then

$$\sigma = \frac{0.213985}{Z(2L_0 + 1)} EFE \frac{4G^2}{1 + R^2} \qquad \text{Mb}$$

where L_0 is total angular momentum quantum number of the bound state. The fits are valid from the threshold up to a maximum frequency ν_{max}. Beyond ν_{max} other methods and data are needed.

12. Because the usual form of the cross section given by Seaton (see *section*2 RBF 4) is non-linear, complicated least squares fitting procedures are needed to obtain the parameters automatically. A similar linear form has already been given in *section*2 RBF 5 and was incorporated into a version of the single channel quantum defect theory program of Peach (1967). Several cross sections were calculated in this way and the fits are generalized so that

$$\ln \sigma = \sum_{i}^{NTERMS} a_i \left[\log\left(\frac{\nu_0}{\nu}\right)\right]^{i-1}.$$

13. Pradhan (1980) recommends that his accurate neon cross sections be linearly interpolated between their values at the various threshold energies. This type deals specifically with that case. The threshold cross sections are held in data statements.

UCOLBB

Types 11 and 12 are intended to be used for fits to the Γ factor as defined by Mihalas (1978 p133). The Γ factor is defined so that the upward collision rate is

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} \Gamma e^{-\frac{h\nu}{kT}}$$

and the fits are respectively

11.

$$\Gamma = \frac{C_{-2}}{T^2} + \frac{C_{-1}}{T} + C_0 + C_1 T$$

and

12.

$$\Gamma = \frac{C_{-2}}{(\ln T)^2} + \frac{C_{-1}}{\ln T} + C_0 + C_1 \ln T.$$

The following (up to and including type 21) are for use with hydrogen and helium although fits of similar form may be suitable in other cases. They are taken from Mihalas and Stone (1968) and from Mihalas (1972).

The close-coupling calculations of Burke *et. al.* (1967) are scaled for Δn ¿ 1 (Δn is the change in the principal quantum number) to the Sampson and Golden (1971) results. The upward rate is

13.

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} \times \left[4 f_{ij} \left(\frac{\nu_H}{\nu_0}\right)^2\right] \left(E_1(u_0) + 0.148 u_0 E_5(u_0)\right)/u_0$$

where $u_0 = e^{-\frac{h\nu_0}{kT}}$ and f_{ij} is the oscillator strength of the corresponding radiative transition. For Δn=1 this is the final result but for Δn ¿ 1, it is scaled further by a factor of

$$\beta + \frac{2(\alpha - \beta)}{\Delta n}$$

in which

$$\alpha = 1.8 - \frac{0.4}{n^2}$$

$$\beta = 3.0 - \frac{1.2}{n}$$

and n is the principal quantum number of the lower level.

14. For He II, the rate is

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} \left(2.71 f_{ij} \left(\frac{\nu_H}{\nu_0}\right)^2\right) u_0 \left[0.693 e^{-\frac{h\nu_0}{kT}} + E_1(u_0)\right]$$

The results are again scaled for Δn ¿ 1, this time by a factor of

$$\min(n, 1.1) \times \min\left[\Delta n, n - \frac{(n-1)}{\Delta n}\right].$$

For allowed transitions from He I ground state Mihalas and Stone (1968) give

15.

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} \left(4 f_{ij} \left(\frac{\nu_H}{\nu_0}\right)^2\right) u_0 E_1(u_0)$$

while from excited states the rate is

16.

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} \left(4 f_{ij} \left(\frac{\nu_H}{\nu_0}\right)^2\right) u_0 \left[E_1(u_0) - e^{-0.2} \frac{u_0}{u_1} E_1(u_1)\right]$$

with $u_1 = u_0 + 0.2$.

The formula A16 of Mihalas and Stone (1968) for the forbidden rates from the ground state of He I is given by type 17. The form is not repeated here as it is not of direct interest. Other forbidden transitions of He I are variously of the form

18.

$$\log \Gamma = C_0 + C_1 \log T + \frac{C_2}{(\log T)^2}$$

and rates

19.

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} a_3 u_0 \left[e^{-u_0} - u_0 E_1(u_0)\right]$$

20.

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} (2a_3) u_0^2 \left[\frac{2}{3}(1 + u_0) e^{-u_0} \right.$$
$$\left. (\pi u_0)^{\frac{1}{2}} (1 - erf(u_0))(1 + \frac{2}{3} u_0)\right]$$

21.

$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} a_3 u_0^2 \left[E_2(u_0) - E_3(u_0)\right]$$

with the error function defined by

$$erf(x) = \int_0^x e^{-t^2} dt.$$

Type 22 is the Van Regemorter formula given previously in *section*2 while type 23 represents fits to some unpublished distorted wave results by Hummer (private communication) for N^{2+} and N^{3+}.

23. If

$$\gamma = \sum_i^{NTERMS} a_i \left(\frac{T}{10^4}\right)^{i-1}$$

the upward rate is

$$\frac{N_e}{T^{\frac{1}{2}}} \frac{g_i}{g_j} e^{-u_0} \gamma$$

with the statistical weights of the two levels g_i, g_j. For optically allowed transitions the rate is further scaled by a factor

$$\frac{f_{obs}}{f_{calc}}.$$

The calculated oscillator strength f_{calc} is the one obtained with the wavefunctions used in the distorted wave scattering problem while 'f_{obs}' is an accurate value, either from an experiment or from a more detailed calculation.

Allen (1973) gives a semi-empirical cross section so that

24.
$$\Gamma = \frac{\Omega}{g_i}\frac{h\nu_H}{kT}.$$

The following are fits to the effective collision strength Υ which is related to the upward collision rate via

$$C_{ij} = \frac{8.631 \times 10^{-6}}{g_j T^{\frac{1}{2}}} N_e e^{-u_0} \Upsilon.$$

25.
$$\Upsilon = \sum_{i=1}^{NFIT} a_i x^{i-1}$$

and x is a scaled temperature

$$x = \frac{T - T1}{scale}.$$

26.
$$\Upsilon = \sum_{i=1}^{NFIT} a_i x^{i-1}$$

where x is now

$$x = \log T - T1.$$

Finally Merts *et. al.* (1980) provide fits for many ions. Their rates have the form

27.
$$8.631 \times 10^{-6} \frac{N_e}{g_i T^{\frac{1}{2}}} u_0 \left[a_1 \frac{e^{-u0}}{u_0} + \left(a_2 + \frac{a_3}{u_0} \right) E_1(u_0) + a_2 E_2(u_0) \right]$$

and

28.
$$\Gamma = \sum_{n=1}^{4} a_n \frac{e^{(u_0 + n a_5)}}{(u_0 + n a_5)}$$

with the upward rate

$$C_{ij} = 8.631 \times 10^{-6} \frac{N_e}{g_i T^{\frac{1}{2}}} u_0 \left(a_0 E_2(u_0) + \Gamma \right)$$

UCOLBF

Again, general fits in powers of T and log T have been included.

11.
$$\Gamma = \frac{C_{-2}}{T^2} + \frac{C_{-1}}{T} + C_0 + C_1 T + C_2 T^2$$

12.
$$\Gamma = \frac{C_{-2}}{(\log T)^2} + \frac{C_1}{\log T} + C_0 + C_1 \log T + C_2 (\log T)^2$$

while a hydrogenic scaling is given by

13.
$$\Gamma = \left(\frac{C_0}{Z} \right)^4.$$

Mihalas and Stone (1968) quote results for He I levels such that the rate is

14.
$$5.465 \times 10^{-11} \frac{N_e}{T^{\frac{1}{2}}} \sigma_0 \left[u_0 E_1(u_0) - 0.728 \frac{u_0^2}{u_1} E_1(u_1) - 0.189 u_0^2 e^{-u_0} \frac{(2+u_2)}{u_2^3} \right]$$

and $u_1 = u_0 + 0.27$, $u_2 = u_0 + 1.43$. They also give the relevant values of σ_0.

The Seaton formula mentioned in *section*2 is repeated as type 15. The Lotz (1968) formula is represented by type 16

16.
$$5.465 \times 10^{-11} \frac{N_e}{T^{\frac{1}{2}}} a_1 \left(2.5 \left(\frac{\nu_H}{\nu_0} \right)^2 \right) u_0 \left[E_1(u_0) - a_2 e^{a_3} \frac{u_0}{u_1} E_1(u_1) \right]$$

with $u_1 = u_0 + a_3$.

Moores (1972) fitted his collisional ionization cross sections for the Boron sequence to a power series in 1/X together with a term in ln X/X. Here X is a dimensionless variable

$$X = \frac{\nu}{\nu_0}.$$

The rate is then

17.
$$5.465 \times 10^{-11} N_e T^{\frac{1}{2}} \left[a_0 E_1(u_0) + a_1 e^{-u_0} + \sum_{i=2}^{N} a_i E_{i-1}(u_0) u_0^{i-1} \right] u_0.$$

The final form in this category are the rates due to Golden and Sampson (1980). These are scaled hydrogenic cross sections that depend on the principal quantum number n, the angular momentum quantum number ℓ, the number of equivalent electrons, $r_{n\ell}$ and a screening factor $\sigma_{n\ell}$ to give an effective charge

$$Z_{eff} = Z - \sigma_{n\ell}.$$

The rate is

18.
$$\begin{aligned} 5.465 \times 10^{-11} \frac{N_e}{T^{\frac{1}{2}}} r_{n\ell} \left(\frac{n^2}{Z_{eff}} \right)^2 & \left[D'(n,\ell) e^{-u_0} \right. \\ & + \left(\tfrac{A'(n,\ell)}{n} + u_0 \left(C'(n,\ell) - Z_{eff} D'(n,\ell) \right) \right) E_1(u_0) \\ & + u_0 \left(D'(n,\ell) + D''(n,\ell) - C'(n,\ell) \right) E_2(u_0) \\ & \left. - D''(n,\ell) E_3(u_0) \right]. \end{aligned}$$

The parameters A′, D′, C′, D″ have been tabulated by Golden and Sampson and are included in the subroutine as a data statement.

UBOUND and UTABLE

These two subroutines are used in conjunction with each other. If type 99 is specified then the keyword **TABLE** should be quoted as indicated in *section*1. This is discussed further below.

Type 11 utilizes the theory of Barnard *et. al.* (1969) for isolated He I lines and is described further by Auer and Mihalas (1972). The depth independent Doppler profiles are again represented by type 12 and the Griem (1960) theory for Stark broadened hydrogenic lines as implemented by Auer and Mihalas (1972) is given by types 13 and 99. If the type is 99, UTABLE is called (providing that **TABLE** has been specified) and the profiles are interpolated from a grid over temperature, density and frequency (see *section*4). If interpolation is not possible, UBOUND will calculate the necessary data since the second

parameter for type 99 (remembering that the oscillator strength is always the first parameter) is a further type (11 – 14) to be used in this eventuality. The data for the second type should then follow as usual.

The last UBOUND type, 14, consists of a Doppler core with power law wings. The width

$$\Delta\nu_S = 8 \times 10^{10} N_e^{\frac{2}{3}} \frac{\lambda(\mathring{A})}{\nu}$$

and

$$x = \frac{\nu - \nu_o}{\Delta\nu_D}.$$

For x ¡ 2, the profile is

$$\frac{1}{\pi^{\frac{1}{2}} \Delta\nu_D} e^{-x^2}$$

while for $x \geq 2$ it is the larger of the above or

$$\left(\frac{C_0}{y}\right)^{2.5}$$

in which $y = \frac{\nu - \nu_0}{\Delta\nu_S}$ and C_0 is a parameter.

C A sample input file

```
:V2
:T================================================================
:T        ALIII  DATASET FOR DETAIL
:T               VERSION 1.03
:T          SET UP BY SYLVIA BECKER
:T================================================================
:V0
:T
:T          EFFECTIVE COLLISION STRENGTHS FOR TRANSITIONS
:T             BETWEEN SOME LEVELS AS IN PAPER BY DUFTON
:T
:T     ***********************************************
:T     *   OLIVER'S  LABELS FOR THE ENERGY-LEVELS    *
:T     ***********************************************
:T
:T
:T----------------------------------------
:T    FREQUENCY MESH
:T----------------------------------------
FREQ
        10      1.E17
        20      6.E16
        30      5.E16
        40      4.E16
        50      3.E16
        60      2.E16
    HHE010      1.3158E16
    HHE011      1.3157E16
        70      1.E16
        80      9.E15
        90      8.E15
     30101      6.87901E15
     30102      6.87871E15
    HHE020      5.9487E15
```

```
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     30201      5.26495E15
     30202      5.26465E15
       100      4.5E15
     30301      3.40274E15
     30302      3.40244E15
    HHE030      3.2929E15
    HHE031      3.2928E15
     30401      3.09675E15
     30402      3.09645E15
     30501      2.57142E15
     30502      2.57112E15
     30601      1.90888E15
     30602      1.90858E15
     30701      1.85412E15
     30702      1.85382E15
     30801      1.76346E15
     30802      1.74316E15
     30901      1.52901E15
     30902      1.52871E15
    HHE040      1.4632E15
    HHE041      1.4631E15
     31001      1.21666E15
     31002      1.21636E15
     31101      1.18669E15
     31102      1.18639E15
     31201      1.18512E15
     31202      1.18482E15
    HHE050      1.1528E15
    HHE051      1.1527E15
    HHE060      9.6034E14
    HHE061      9.6033E14
    HHE070      8.7612E14
    HHE071      8.7611E14
    HHE080      8.2238E14
    HHE081      8.2237E14
    HHE090      8.1472E14
    HHE091      8.1471E14
       110      6.5E14
    HHE100      5.2631E14
    HHE101      5.2630E14
    HHE110      3.6548E14
    HHE111      3.6547E14
    HHE120      2.6852E14
    HHE121      2.6851E14
    HHE130      2.0551E14
    HHE131      2.0550E14
    HHE140      1.6244E14
    HHE141      1.6243E14
    HHE150      1.3153E14
    HHE151      1.3152E14
       120      8.E13
       130      5.E13
       140      1.E13
       150      5.E12
       160      1.E12
       170      1.E11
 0
:V2
:T-------------------------------------------
```

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:T-------------------------------------------
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L
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  AL32P1     6.      5.26480E15        AL41S1
  AL32D1    10.      3.40259E15        AL41S1
  AL32S2     2.      3.09660E15        AL41S1
  AL32P2     6.      2.57127E15        AL41S1
  AL32D2    10.      1.90873E15        AL41S1
  AL32F1    14.      1.85397E15        AL41S1
  AL32S3     2.      1.76331E15        AL41S1
  AL32P3     6.      1.52886E15        AL41S1
  AL32D3    10.      1.21651E15        AL41S1
  AL32F2    14.      1.18654E15        AL41S1
  AL32G1    18.      1.18497E15        AL41S1

0
S
  3NS2S     2.     0.    2.961E16   0.9        6   10    AL41S1
  3NP2P     6.     0.    2.961E16   0.6        6   10    AL41S1
  3ND2D    10.     0.    2.961E16   0.07       6   10    AL41S1
  3NF2F    14.     0.    2.961E16   0.007      6   10    AL41S1
  3NG2G    18.     0.    2.961E16   0.0025     6   10    AL41S1
  3NH2H    22.     0.    2.961E16   0.0020     6   10    AL41S1
  3REST   -72.     2.    2.961E16   0.0        7   10    AL41S1
0
K
  AL41S1    1.0      2.90134E16       NONE
:V2
:T-------------------------------------------------
:T ALUMINIUM STATISTICS AND OPACITIES
:T-------------------------------------------------
TL   30000.
DB 9    -4.    -2.    -1.33   -.66   0.   .66    1.33    2.   4.
TY   RBB     1    1    RBB
   AL32S1     AL32P1     .854
   AL32S1     AL32P2     .0133
   AL32S1     AL32P3     .0076
   AL32P1     AL32D1     .928
   AL32P1     AL32S2     .127
   AL32P1     AL32D2     .0039
   AL32P1     AL32S3     .0180
   AL32P1     AL32D3     .00005
   AL32D1     AL32P2     .171
   AL32D1     AL32F1     .936
   AL32D1     AL32P3     .0097
   AL32D1     AL32F2     .166
   AL32S2     AL32P2    1.28
   AL32S2     AL32P3     .0062
   AL32P2     AL32D2    1.28
   AL32P2     AL32S3     .229
   AL32P2     AL32D3     .0229
   AL32D2     AL32P3     .345
   AL32D2     AL32F2     .732
   AL32F1     AL32D3     .0187
   AL32F1     AL32G1    1.339
   AL32S3     AL32P3    1.660
0
TY   RBF    12   6
```

```
   RBF AL41S1  AL32S1  10  30101 -42.363   .1666    -.90357   -.18621  .0  .0
               AL32P1  10  30201 -42.606  3.95E-2   -.84268   -.14762  .0  .0
               AL32D1  10  30301 -40.468  3.0378    -.17911   -.1278   .0  .0
               AL32S2  10  30401 -42.268   .68105   -.44824   -6.37E-2 .0  .0
               AL32P2  10  30501 -42.241  1.3642     .22185    .11694  .0  .0
               AL32D2  10  30601 -39.952  2.8394    2.803E-2  8.471E-2  .0 .0
               AL32S3  10  30801 -42.132   .69576   -.40463   -5.142E-2 .0 .0
               AL32P3  10  30901 -41.690  2.0366    .55522     .13478   .0 .0
               AL32D3  10  31001 -39.560  2.3479   -.20798    -8.091E-3 .0 .0
0
TY   RBF     5    1
  RBF    AL41S1    AL32F1     10   30701    4
                   AL32F2     10   31101    5
                   AL32G1     10   31201    5
  0
TY   CBB    22    2    CBB
   AL32S1     AL32P3     .0076     .2
   AL32P1     AL32D2     .0039     .2
   AL32P1     AL32S3     .0180     .2
   AL32P1     AL32D3     .00005    .2
   AL32D1     AL32F1     .936      .2
   AL32D1     AL32P3     .0097     .2
   AL32D1     AL32F2     .166      .2
   AL32S2     AL32P3     .0062     .2
   AL32P2     AL32D2    1.28       .7
   AL32P2     AL32S3     .229      .2
   AL32P2     AL32D3     .0229     .2
   AL32D2     AL32P3     .345      .2
   AL32D2     AL32F2     .732      .2
   AL32F1     AL32D3     .0187     .2
   AL32F1     AL32G1    1.339      .2
   AL32S3     AL32P3    1.660      .7
0
TY   CBB   24    1  CBB
 AL32S1  AL32D2  1.0 AL32S1  AL32F1  1.0 AL32S1  AL32S3  1.0
 AL32S1  AL32D3  1.0 AL32S1  AL32F2  1.0 AL32S1  AL32G1  1.0
 AL32P1  AL32F1  1.0
 AL32P1  AL32P3  1.0 AL32P1  AL32F2  1.0
 AL32P1  AL32G1  1.0 AL32D1  AL32D2  1.0
 AL32D1  AL32S3  1.0 AL32D1  AL32D3  1.0
 AL32D1  AL32G1  1.0 AL32S2  AL32D2  1.0
 AL32S2  AL32F1  1.0 AL32S2  AL32S3  1.0 AL32S2  AL32D3  1.0
 AL32S2  AL32F2  1.0 AL32S2  AL32G1  1.0 AL32P2  AL32F1  1.0
 AL32P2  AL32P3  1.0 AL32P2  AL32F2  1.0
 AL32P2  AL32G1  1.0 AL32D2  AL32F1  1.0 AL32D2  AL32S3  1.0
 AL32D2  AL32D3  1.0 AL32D2  AL32G1  1.0 AL32F1  AL32S3  1.0
 AL32F1  AL32P3  1.0 AL32F1  AL32D3  1.0 AL32F1  AL32F2  1.0 AL32F1  AL32G1  1.0
 AL32S3  AL32P3  1.0 AL32S3  AL32D3  1.0 AL32S3  AL32F2  1.0 AL32S3  AL32G1  1.0
 AL32P3  AL32D3  1.0 AL32P3  AL32F2  1.0 AL32P3  AL32G1  1.0 AL32D3  AL32F2  1.0
 AL32D3  AL32G1  1.0 AL32F2  AL32G1  1.0
0
TY    CBF    15     1
   CBF  AL41S1    AL32S1     4.E-19
                  AL32P1     3.15E-19
                  AL32D1     3.67E-18
                  AL32S2     4.35E-19
                  AL32P2     4.73E-19
                  AL32D2     4.31E-18
                  AL32S3     5.E-19
                  AL32P3     3.32E-19
```

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                 AL32D3     6.54E-18
                 AL32F1     3.16E-17
                 AL32F2     3.96E-17
                 AL32G1     3.96E-17
   0
:V2
TY  CBB   26  5  CBB
     AL32S1   AL32P1   4.75   3   20.2375      8.5905      4.0476
     AL32S1   AL32D1   4.75   3   3.92094      0.239048    0.244048
     AL32S1   AL32S2   4.75   3   1.09469     -.0209524    0.109127
     AL32S1   AL32P2   4.75   3   0.620625    -.00190476   -.0277777
     AL32P1   AL32D1   4.75   3   45.7625     22.5238      13.4127
     AL32P1   AL32S2   4.75   3   3.27719     1.11429      1.64881
     AL32P1   AL32P2   4.75   3   5.40250     0.490476     0.238095
     AL32D1   AL32S2   4.75   3   10.1181     -3.80667     0.115079
     AL32D1   AL32P2   4.75   3   38.0187     4.63810      -1.78571
0
TY  CBB   26  6  CBB
     AL32S2   AL32P2   4.75   4   79.8375   31.7088   21.8254   5.41838
0
:V0
:T ----------------------------------------------------------------
:T HYDROGEN BACKGROUND DETAIL DATASET  VERSION 100 RELEASED 13/11/85
:T ----------------------------------------------------------------
 ATOM H 0.0 1.008 0.91
 L
H11       2.     0.328805282E 16   H21
H12       8.     0.822013206E 15   H21
H13      18.     0.365339202E 15   H21
H14      32.     0.205503301E 15   H21
H15      50.     0.131522113E 15   H21
 0
 K
H21        1.      0.0             NONE
:T PHOTOIONIZATIONS FROM H I TO H II
 TY RBF 5 1 RBF H21
      H11  10 HHE030  1
      H12  10 HHE080  2
      H13  10 HHE110  3
      H14  10 HHE130  4
      H15  10 HHE150  5
 0
:T FREE FREE ABOVE H I
 TY RFF 1 0 RFF H21  0    TY RFF 2 1 RFF H21  (6)   0
:T ......END OF HYDROGEN BACKGROUND DATASET...............
:V0
:T ----------------------------------------------------------------
:T HELIUM BACKGROUND DETAIL DATASET  VERSION 100 RELEASED 13/11/85
:T VERSION 100 SIMULATES HELIUM LINE BLOCKING 228-261 A.
:T ----------------------------------------------------------------
 ATOM HE 0.0 4.0026 0.09  L
HE11        1.     5.948620002E15  HE21
HE12S1      1.     9.603359103E14  HE21
HE12P1      3.     8.147174578E14  HE21
HE12S3      3.     1.152845000E15  HE21
HE12P3      9.     8.761150639E14  HE21
HE13       36.     3.65339202E 14  HE21
HE14       64.     2.05503301E 14  HE21
HE15      100.     1.31522113E 14  HE21
 0
```

```
 L
HE21      2.     0.131575659E 17   HE31
HE22      8.     0.328939147E 16   HE31
HE23     18.     0.146195176E 16   HE31
HE24     32.     0.822347868E 15   HE31
HE25     50.     0.526302636E 15   HE31
HE26     72.     0.365487941E 15   HE31
HE27     98.     0.268521753E 15   HE31
HE28    128.     0.205586967E 15   HE31
HE29    162.     0.162439085E 15   HE31
HE210   200.     0.131575659E 15   HE31
 0
 K
HE31        1.     0.0            NONE
:T PHOTOIONIZATIONS FROM HE I TO HE II
 TY RBF 6 3 RBF HE21  HE11   10 HHE020 (7.3E-18 1.373E0 -2.311E-16)      0
 TY RBF 2 2 RBF HE21  HE12S1 10 HHE060 (1.079E-17 -1.91)                 0
 TY RBF 3 4 RBF HE21  HE12P1 10 HHE090  (1.322E-17 -3.5 6.315E-19 -3.3)
                      HE12P3 10 HHE070  (1.726E-17 -2.9 5.627E-19 -3.3)  0
 TY RBF 7 3 RBF HE21  HE12S3 10 HHE050  (-2.783E2 1.488E1 -2.311E-1)     0
 TY RBF 5 1 RBF HE21  HE13   10 HHE110  3
                      HE14   10 HHE130  4
                      HE15   10 HHE150  5                                0
:T FREE FREE ABOVE HE I
 TY RFF 1 0 RFF HE21  0    TY RFF 2 1  RFF HE21 (6)   0
:T PHOTOIONIZATIONS FROM HE II TO HE III
 TY RBF 5 1 RBF HE31  HE21  10 HHE010   1
                      HE22  10 HHE030   2
                      HE23  10 HHE040   3
                      HE24  10 HHE080   4
                      HE25  10 HHE100   5
                      HE26  10 HHE110   6
                      HE27  10 HHE120   7
                      HE28  10 HHE130   8
                      HE29  10 HHE140   9
                      HE210 10 HHE150  10
 0
:T FREE FREE ABOVE HE II
 TY RFF 1 0 RFF HE31  0    TY RFF 2 1 RFF HE31 (11)    0
:T ..........END OF HELIUM BACKGROUND DATASET...............
MULTI 1
:V2
LINEAR 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
27 28 29 30 31 32 33 34
      0
```

D A sample output file

```
 CONTINUUM FREQUENCY WINDOW

  IJ  LABEL        FRQ(HZ)             WAVE(A)

   1  1           1.0000000E+17        29.97925
   2  FA          8.0000000E+16        37.47406
   3  FB          6.0000000E+16        49.96542
   4  2           5.0000000E+16        59.95850
   5  01          4.0000000E+16        74.94812

 ATOM   1 HE     ZEFF   0.00000  WT   4.00260 ABUNDANCE  0.090000
```

L	LABEL	G	FION	PARENT
1	HE11S1	1.	5.9486200E+15	HE21
2	HE12S1	1.	9.6033591E+14	HE21
3	HE12P1	3.	8.1471746E+14	HE21
4	HE13S1	1.	4.0310167E+14	HE21
5	HE13P1	3.	3.6279150E+14	HE21
6	HE13D1	5.	3.6592734E+14	HE21
7	HE14S1	1.	2.2096243E+14	HE21
8	HE14P1	3.	2.0440037E+14	HE21
9	HE14D1	5.	2.0578661E+14	HE21
10	HE14F1	7.	2.0559084E+14	HE21
11	HE15N1	25.	1.3163600E+14	HE21
12	HE16N1	36.	9.1413889E+13	HE21
13	HE17N1	49.	6.7161224E+13	HE21
14	HE18N1	64.	5.1420313E+13	HE21
15	HE12S3	3.	1.1528450E+15	HE21
16	HE12P3	9.	8.7611506E+14	HE21
17	HE13S3	3.	4.5190497E+14	HE21
18	HE13P3	9.	3.8211616E+14	HE21
19	HE13D3	15.	3.6604725E+14	HE21
20	HE14S3	3.	2.4021034E+14	HE21
21	HE14P3	9.	2.1266116E+14	HE21
22	HE14D3	15.	2.0584387E+14	HE21
23	HE14F3	21.	2.0561003E+14	HE21
24	HE15N3	75.	1.3160800E+14	HE21
25	HE16N3	108.	9.1394444E+13	HE21
26	HE17N3	147.	6.7146939E+13	HE21
27	HE18N3	192.	5.1409375E+13	HE21

S	LABEL	GS	RYD	PARENT	NS
1	HE19N1	0.00	3.2909000E+15	HE21	9
		1.00			16
2	HE19N3	0.00	3.2902000E+15	HE21	9
		3.00			16

L	LABEL	G	FION	PARENT
28	HE21	2.	1.3157566E+16	HE31
29	HE22	8.	3.2893915E+15	HE31
30	HE23	18.	1.4619518E+15	HE31
31	HE24	32.	8.2234787E+14	HE31
32	HE25	50.	5.2630264E+14	HE31
33	HE26	72.	3.6548794E+14	HE31
34	HE27	98.	2.6852175E+14	HE31
35	HE28	128.	2.0558697E+14	HE31
36	HE29	162.	1.6243908E+14	HE31
37	HE210	200.	1.3157566E+14	HE31
38	HE211	242.	1.0874021E+14	HE31
39	HE212	288.	9.1371985E+13	HE31
40	HE213	338.	7.7855419E+13	HE31
41	HE214	392.	6.7130438E+13	HE31

X	LABEL	G	FION	PARENT
1	HE215	450.	5.8478071E+13	HE31
2	HE216	512.	5.1396742E+13	HE31

S	LABEL	GS	RYD	PARENT	NS

```
  3 HE217     0.00    1.3157566E+16     HE31      17
              2.00                                32

 K  LABEL     G         FION            PARENT

 42 HE31      1.      0.0000000E+00     HE31
TL       30000.00
DB      9    -4.8000   -3.2000   -1.6000   -0.8000    0.0000    0.8000    1.6000
*  OPTICALLY FORBIDDEN COLLISIONS IN SINGLET HE I

TY   CBB       35     2
  1 CBB     1 HE      1 HE11S1 15 HE12S3  0        TY 35  2     WAVE     625.1179
                                        DATA    1.000000       2.000000
  2 CBB     1 HE      1 HE11S1  2 HE12S1  0        TY 35  2     WAVE     600.9932
                                        DATA    1.000000       3.000000
  3 CBB     1 HE      1 HE11S1 16 HE12P3  0        TY 35  2     WAVE     591.0147
                                        DATA    1.000000       4.000000
*  COLLISIONAL IONIZATION FROM HE I SINGLETS TO HE II

TY   CBF       14     1
112 CBF     1 HE      1 HE11S1 28 HE21    0        TY 14  1     WAVE     503.9698
                                        DATA    1.640000
113 CBF     1 HE      2 HE12S1 28 HE21    0        TY 14  1     WAVE     3121.746
                                        DATA    22.90000
114 CBF     1 HE      3 HE12P1 28 HE21    0        TY 14  1     WAVE     3679.711
                                        DATA    20.40000
115 CBF     1 HE      4 HE13S1 28 HE21    0        TY 14  1     WAVE     7437.144

TY   RBB        1     1
  1 RBB     1 HE      1 HE11S1  3 HE12P1  0        TY  1  1     WAVE     583.9466
                                        DATA   5.1339025E+15  0.2811000
  2 RBB     1 HE      2 HE12S1  3 HE12P1  0        TY  1  1     WAVE     20587.54
                                        DATA   1.4561845E+14  0.3621000
  3 RBB     1 HE      3 HE12P1  4 HE13S1  0        TY  1  1     WAVE     7283.309
                                        DATA   4.1161579E+14  4.6700000E-02

TY   RBF        6     3
126 RBF     1 HE      1 HE11S1 28 HE21    0        TY  6  3     WAVE     503.9698
                                        DATA   7.3000000E-18   1.373000     -2.3110000E-16
TY   RBF        7     3
127 RBF     1 HE     15 HE12S3 28 HE21    0        TY  7  3     WAVE     2600.458
                                        DATA   -278.3000       14.88000     -0.2311000

TY   RFF        1     0
  1 RFF     1 HE     28 HE18N3 28 HE21    0        TY  1  0     WAVE    0.0000000E+00
TY   RFF        2     1
  2 RFF     1 HE     28 HE18N3 28 HE21    0        TY  2  1     WAVE    0.0000000E+00
                                        DATA    9.000000

MULTI  2


*   HE11S1-HE12P1 HE12S1-HE12P1 HE12P1-HE13S1 HE12P1-HE13D1 HE11S1-HE13P1

LINEAR   1 RBB     1 HE      1 HE11S1  3 HE12P1
LINEAR   2 RBB     1 HE      2 HE12S1  3 HE12P1
LINEAR   3 RBB     1 HE      3 HE12P1  4 HE13S1
LINEAR   4 RBB     1 HE      3 HE12P1  6 HE13D1
LINEAR   5 RBB     1 HE      1 HE11S1  5 HE13P1
```

1BACKGROUND CHANNELS

CHANNEL	ATOM	LOWER	UPPER	VIA	IFL	IFU	WAVELENGTH	
7 RBB	1 HE	4 HE13S1	5 HE13P1	0	2558	2566	74358.20	74362.61
							74371.44	74373.64
							74384.68	
8 RBB	1 HE	6 HE13D1	5 HE13P1	0	2907	2915	955851.1	955907.8
							956021.3	956049.7
							956191.5	
127 RBF	1 HE	15 HE12S3	28 HE21	0	1	520	29.97925	2600.431
128 RBF	1 HE	2 HE12S1	28 HE21	0	1	578	29.97925	3121.714
129 RBF	1 HE	16 HE12P3	28 HE21	0	1	626	29.97925	3421.803

1LINEARIZED CHANNELS

CHANNEL	ATOM	LOWER	UPPER	VIA	IFL	IFU	WAVELENGTH	
1 RBB	1 HE	1 HE11S1	3 HE12P1	0	78	86	583.8427	583.8773
							583.9466	583.9639
							584.0506	
2 RBB	1 HE	2 HE12S1	3 HE12P1	0	1931	1939	20583.87	20585.09
							20587.54	20588.15
							20591.20	
3 RBB	1 HE	3 HE12P1	4 HE13S1	0	1134	1142	7282.012	7282.445
							7283.309	7283.525
							7284.606	
36 RBF	1 HE	1 HE11S1	28 HE21	0	1	20	29.97925	503.9644
37 RBB	1 HE	29 HE22	35 HE28	0	256	267	971.9784	972.0361
							972.1226	972.1514
							972.2091	972.2669

MODEL

```
TITLE HUSFELD/GROTH MODEL ATMOSPHERE PROGRAM
TITLE MODEL STARTED ON1991-09-03 AT  21:04:32
TITLE NLTE-LEVELS IN H/HE1/HE2=5/5/10 , LINE-MODEL
TITLE STATUS:CONV AFTER   14 ITERATIONS
TEFF      42500.0
GRAVITY    3.5500
YHE        0.1401
ABUNDANCE   H         0.8598450
ABUNDANCE   HE        0.1401550
```

STRUCTURE 90 TITLE (OUTPUT FROM MOD2AM)

ID	MASS	TEMP	NA	NE
1	3.1325500E-07	36579.3	4.7059700E+07	5.3655300E+07
2	9.9059400E-07	36578.6	1.4881800E+08	1.6967500E+08
3	3.1325000E-06	36576.3	4.7063100E+08	5.3658200E+08
4	3.9435800E-06	36575.5	5.9250300E+08	6.7552800E+08
5	4.9646500E-06	36574.4	7.4593700E+08	8.5045800E+08
6	6.2501000E-06	36573.0	9.3911300E+08	1.0706900E+09

POPULATIONS 90 TITLE NON-LTE H AND HE FROM MODEL

ATOM 2 H ZEFF 0.00000 WT 1.00800 ABUNDANCE 0.859845

L	LABEL	G	FION	PARENT

```
  1 H11       2.      3.2900000E+15     H21
 1.96426E-05 1.96436E-04 1.96471E-03 3.11407E-03 4.93590E-03 7.82374E-03 1.24015E-02 1.96586
 7.83309E-02 1.24204E-01 1.96968E-01 3.12417E-01 4.95655E-01 7.86618E-01 1.24892E+00 1.98402
 7.98538E+00 1.27140E+01 2.02403E+01 3.22054E+01 5.12043E+01 8.13304E+01 1.29014E+02 2.04332
 5.71337E+02 8.03375E+02 1.12874E+03 1.58471E+03 2.22303E+03 3.11511E+03 4.35903E+03 6.08936
 1.63987E+04 2.27082E+04 3.13622E+04 4.31919E+04 5.92945E+04 8.10960E+04 1.10435E+05 1.49662
 3.59823E+05 4.75762E+05 6.24434E+05 8.12957E+05 1.17876E+06 1.67376E+06 2.32327E+06 3.14582
 6.66000E+06 8.14790E+06 9.78617E+06 1.15923E+07 1.36118E+07 1.59273E+07 1.86622E+07 2.19791
 3.75308E+07 4.54565E+07 5.53140E+07 6.75512E+07 8.27029E+07 1.01349E+08 1.24002E+08 1.50943
 2.56416E+08 3.00342E+08 3.51655E+08 4.15346E+08 4.99339E+08 6.14933E+08 7.77457E+08 1.00737
POPULATIONS END

TURBULENCE   0.00


PHYSICAL SETUP
   RUN STATISTICS FULL PART
      ATOMS          2    2
      LEVELS        53   53
      LTE LEVELS     8
      Q.D. SERIES    3
      COLLISIONS   451
      FREQUENCIES 2947 1134
      OPACITIES    325
      RATES        325   98
      FREE FREE      6
      CROSS DATA  1522 1304  218
      INPUT DATA  1300
      DEPTHS        90

   PRINT PARAMETERS

      POPULATIONS 90  COLLISIONS 90  CROSS DATA 90  OPACITY 90

ATOMIC DATA TITLE :DETAIL DATASET FOR HE I  VERSION 1.02 FROM K.B.  MORE LINEAR

MODEL TITLE       :HUSFELD/GROTH MODEL ATMOSPHERE PROGRAM
MODEL TITLE       :MODEL STARTED ON1991-09-03 AT  21:04:32
MODEL TITLE       :NLTE-LEVELS IN H/HE1/HE2=5/5/10 , LINE-MODEL
MODEL TITLE       :STATUS:CONV AFTER   14 ITERATIONS

DETAIL PROGRAM VERSION  2.00 ; EXECUTION DATE 08/25/93    TIME 17:40:47

MODEL PARAMETERS  :TEFF     42500.
                   GRAVITY      3.550
                   NHE/NA       0.1401
                   TLINE    30000.
                   ABUNDANCES         1    HE              0.1402
                                      2    H               0.8598
                   TURBULENCE         1     0.000     0.000

                   ALBEDO             1        0.000000E+00   0.00000


FLOATING ATOMS   :
                     1 HE     T          2 H      T

FLOATING LEVELS  :
                     1 HE11S1 T          2 HE12S1 T          3 HE12P1 T
                     4 HE13S1 T          5 HE13P1 T          6 HE13D1 T
```

```
                    7 HE14S1 T          8 HE14P1 T          9 HE14D1 T
                   10 HE14F1 T         11 HE15N1 T         12 HE16N1 T
                   13 HE17N1 T         14 HE18N1 T         15 HE12S3 T
                   16 HE12P3 T         17 HE13S3 T         18 HE13P3 T
                   19 HE13D3 T         20 HE14S3 T         21 HE14P3 T
                   22 HE14D3 T         23 HE14F3 T         24 HE15N3 T
                   25 HE16N3 T         26 HE17N3 T         27 HE18N3 T
                   28 HE21   T         29 HE22   T         30 HE23   T
                   31 HE24   T         32 HE25   T         33 HE26   T
                   34 HE27   T         35 HE28   T         36 HE29   T
                   37 HE210  T         38 HE211  T         39 HE212  T
                   40 HE213  T         41 HE214  T         42 HE31   T
                   43 H11    T         44 H12    T         45 H13    T
                   46 H14    T         47 H15    T         48 H16    T
                   49 H17    T         50 H18    T         51 H19    T
                   52 H110   T         53 H21    T
DEPTH   1 SUMUP POPS
 7.37694E-11 4.84911E-13 1.45473E-12 1.38480E-13 3.94038E-13
 6.59438E-13 1.09044E-13 3.20100E-13 5.34472E-13 7.48068E-13
 3.44309E-12 3.31197E-12 4.36678E-12 5.58696E-12 1.45473E-12
 4.36420E-12 4.42910E-13 1.21247E-12 1.97862E-12 3.35499E-13
 9.70766E-13 1.60354E-12 2.24426E-12 1.03293E-11 9.93567E-12
 1.31001E-11 1.67607E-11 1.02771E+02 8.11904E-06 2.44100E-06
 1.99563E-06 1.97614E-06 2.14450E-06 2.39636E-06 2.72262E-06
 3.10393E-06 3.53153E-06 2.92336E-06 3.40066E-06 3.92090E-06
 4.48378E-06 6.59555E+06 1.96426E-05 1.03759E-06 1.22323E-06
 1.86147E-06 5.42198E-06 5.21556E-06 6.87682E-06 8.79853E-06
 1.09792E-05 1.34181E-05 4.04641E+07
DEPTH   1 SAHA FACTORS
 1.94708E-12 2.79918E-15 6.93710E-15 1.34746E-15 3.83414E-15
 6.41657E-15 1.06104E-15 3.11469E-15 5.20061E-15 7.27898E-15
 2.35924E-14 3.22267E-14 4.24904E-14 5.43632E-14 1.08105E-14
 2.25571E-14 4.30968E-15 1.17978E-14 1.92528E-14 3.26453E-15
 9.44591E-15 1.56030E-14 2.18375E-14 7.07747E-14 9.66778E-14
 1.27469E-13 1.63087E-13 9.98027E-08 9.51174E-13 1.94606E-13
 1.49480E-13 1.58386E-13 1.84691E-13 2.21354E-13 2.66201E-13
 3.18368E-13 3.77449E-13 4.43233E-13 5.15600E-13 5.94477E-13
 6.79819E-13 1.00000E+00 2.37387E-13 3.73541E-14 4.61638E-14
 6.65431E-14 9.43557E-14 1.28894E-13 1.69949E-13 2.17440E-13
 2.71332E-13 3.31605E-13 1.00000E+00

UPWARD COLLISIONS  1

 LOWER UPPER   CU        LOWER UPPER   CU        LOWER UPPER   CU        LOWER UPPER   CU

HE11S1HE12S3 3.150E-04  HE11S1HE12S1 1.563E-04  HE11S1HE12P3 1.197E-04  HE11S1HE12P1 9.316E-
HE11S1HE13S1 2.145E-05  HE11S1HE13P3 2.249E-05  HE11S1HE13D3 4.804E-06  HE11S1HE13D1 1.109E-
HE12S3HE12S1 1.241E+00  HE12S3HE12P3 3.289E+01  HE12S3HE12P1 5.645E-01  HE12S3HE13S3 8.526E-
HE12S3HE13P3 6.077E-01  HE12S3HE13D3 1.250E+00  HE12S3HE13D1 1.015E-01  HE12S3HE13P1 5.267E-
HE12S1HE12P1 7.780E+01  HE12S1HE13S3 4.597E-01  HE12S1HE13S1 1.052E+00  HE12S1HE13P3 4.785E-

CROSS SECTIONS   1   2 RFF    28 HE21   28 HE21      1 2947
 5.666E-39 1.107E-38 2.623E-38 4.533E-38 8.854E-38 1.426E-37 2.296E-37 3.697E-37 5.954E-37 9
 2.486E-36 2.488E-36 2.488E-36 4.004E-36 6.447E-36 1.038E-35 1.672E-35 2.692E-35 2.692E-35 2
 2.762E-35 2.762E-35 2.763E-35 2.763E-35 2.763E-35 2.763E-35 2.764E-35 2.764E-35 2.784E-35 2
 2.785E-35 2.785E-35 2.786E-35 2.786E-35 2.787E-35 2.818E-35 2.819E-35 2.819E-35 2.820E-35 2
 2.821E-35 2.821E-35 2.877E-35 2.878E-35 2.878E-35 2.878E-35 2.879E-35 2.879E-35 2.879E-35 2

FIXED OPACITY  1
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    1 3.5689E-17   1 8.0489E-87   2 3.5689E-17   2 3.9148E-75   3 3.5689E-17   3 2.3112E-63
    5 3.5689E-17   5 1.9437E-51   6 3.5689E-17   6 6.9607E-48   7 3.5689E-17   7 8.0395E-45
    9 3.5689E-17   9 6.8264E-40  10 3.5689E-17  10 6.5265E-38  11 3.5689E-17  11 3.4266E-36
   13 3.5689E-17  13 1.0793E-34  14 3.5689E-17  14 1.0831E-34  15 3.5689E-17  15 1.0834E-34

 CROSS SECTIONS   1   1 RBB     1 HE11S1  3 HE12P1   78   86
  1.244E-22 4.509E-18 2.452E-15 1.184E-14 2.001E-14 1.184E-14 2.452E-15 4.509E-18 1.244E-22

 CROSS SECTIONS   1 126 RBF     1 HE11S1 28 HE21      1   20
  2.649E-27 2.694E-25 2.740E-23 2.763E-22 2.786E-21 1.083E-20 3.447E-20 9.259E-20 2.151E-19 4
  1.377E-18 1.377E-18 1.377E-18 2.152E-18 3.149E-18 4.359E-18 5.752E-18 7.287E-18

1MODEL STRUCTURE

 ID   MASS              TEMP        NA             NE             NE SUM

  1 3.132550E-07       36579.30   4.705970E+07   5.365530E+07   5.365530E+07
  2 9.905940E-07       36578.60   1.488180E+08   1.696750E+08   1.696752E+08
  3 3.132500E-06       36576.30   4.706310E+08   5.365820E+08   5.365821E+08
  4 3.943580E-06       36575.50   5.925030E+08   6.755280E+08   6.755279E+08

 PRINT    3  90
 DRIVE
1SURFACE FLUX

   IJ      FREQUENCE WAVELENGTH           HNU  M(TAU=1)    TAUMAX    ALBEDO

     1    1.0000000E+17        29.97925     2.526E-45 6.512E-02 1.861E+02     0.000
     2    8.0000000E+16        37.47406     2.839E-45 4.775E-02 2.314E+02     0.000
     3    6.0000000E+16        49.96542     5.715E-34 3.077E-02 3.605E+02     0.000
     4    5.0000000E+16        59.95850     3.131E-28 2.162E-02 5.218E+02     0.000
     5    4.0000000E+16        74.94812     1.328E-22 1.286E-02 8.740E+02     0.000
1LOG CHANGES IN POPULATIONS

0ID -8        -7        -6        -5        -4        -3        -2        -1         0
  1 .         .  -42    -53       .         .         .       30428 -4144-5283-475829
  2 .         .         .  -4-53  .         .         .       30428 -4144-5283-475829
  3 .         .   -42-53.         .         .         .       30428 -4144-5283-474829
  4 .         .    -42  .53       .         .         .       30428 -4144-5283-474829
  5 .         .        -53  -42   .         .         .       30428 -414-52163-474829
  6 .         .    -53  .         .         .         .       30428 -410-52163-474829
  7 .         .      53-42        .         .         .       30428 -410-52063-474829
  8 .         .         -453      .         .         .       30428 -410-52063-474829
  9 .         .         .   532   .         .         .       30428 -41-521063-474829
 10 .         .         .   -53 -42         .         .       30428 -41-521946-474829
 11 .         .         .-53      -42       .         .       30428 -41-521946-474829
 12 .         .  53     .         -42       .         .       30428 -4-5210946-474829
 13 .         .53       .         .-42      .         .       30-28-41-5210946-474829
 14 .         .      53 .         . -42     .         .        3028-4-52510846-474829
 15 .         . -53     .         .  -42    .         .        3028--525150846-474829
 16 .         -53       .         .   -42   .         .        3028--525109846-474829
 17 .         .        53         .    -42  .         .        308--52-5109846-474829
 18 .         .         .-53      .     -42 .         .         30--5251509846-474829
 19 .         .         . 53      .     -42 .         .         30-52-5104944-4754829
 20 .         .        53         .      -42.         .         3-52-51509844-4754829
 21 .         .         . -53     .       -42         .        2-52-515049844-4753829
 22 .         .         53        .       -42         .        2-52-515049844-4753529
 23 .         .         -53       .       -42         .       2-52-51-5094844-4753529
 24 .         .        53         .      -42.         .      2-524-515049484-47358529
 25 .         .     -53 .         .     -42 .         .     2-524-5150-49834-47548529
```

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26 .       53.         .         .     -42 .          .    28-52-51-5049483-476548529
27 .         .         53        .   -42   .          .  28 -52-51-50-49483-473548529
28 .         .         53        .  -42    .          . 28 -524-51-5049484-4763548529
29 .         .    -53  .         .-42      .         28  -524-51-50-49484-4765438529
30 .         .    53   .         -42       .        28   -52-51-50-49-48-4746543829.
31 .         .         -53      -42        .      28 .24-524-51-50-49483-4765543829.
32 .         .       -53      -42.         .     28  .  -52-51-50-49-48-47465343829.
33 .         .         .-53 -42  .         .    28   . -524-51-50-49483-47465431829.
34 .         .         . -5342   .         .   28    .-524-51-50-49-48-474465431829.
35 .         .      53 .  -42    .         .   28    -524-51-50-49-4877-46453431829.
36 .         .         53-42     .         .   28    -521-51-50-49-48-4744464301829.
37 .         .     -53 .  -42    .         .   28   -521-51-50-49-48-4734465430129 .
38 .         .   -53   .    -42  .         .   28  -52--51-508-49-48-4764645311129 .
39 .         .      -53-42       .         .   28 --524-51-50 -49-48--474446301815 .
40 .         .      -42.      53 .         .   --524-513-50-49-4837466-44543301215 .
41 .         .        42     53  .         -41--529-51-50--4948-37466-345433301215 .
42 .         .         53   -42  41      40. -52 -518-50-4948 -37-36-3445323302915 .
43 .         .       -53 -42     .         -522-51-50498    -3747-4634543232302915 .
44 .         .    53   .   -42   .         .  52  51 35034497  486--4544713-469715 .
45 .         .         .-53 -42  .       -52 -51-50-49-48-473-36-34644454130308615 .
46 .         .      53 .   -42   .   -52-51-50-491 4-479 38-36 -346222453113098615 .
47 .         .         53 -4252  .51   50  -449341  4839 386 -3544623454231308715  .
48 .    53   .         .  -42    .   52   51850 44947048638354 . 4623454313098115  .
49 .         .      53 .    -42  .      52 251 501649047488    346-24542313087115  .
50 .         .         .53   -42 .    35 522851 50344934488    463244543310988115  .
51 .         .        -53      -42       522851 450449394484   463245443101985115  .
52 .         .      -53.       -42       52285141504049944834 4633 45423191971615  .
53 .         .         .  -53    -42     528 514150049934483446.3345443104195115   .
54 .         .  -53    .        -42      528 511 500499348 346 33454431041191615   .
55 .         .         -53       -42    528. 511500 493348346 33454431931197615    .
56 .         .         .     53  -42    528.511 500493348346 33454410223 191615    .
57 .         .         . 53     -42    528 511 50049 34834633454431012331976 15     .
58 .         .       -53        -42   5228511 500 4934486 33454430752343196 12     .
59 .         .        53       -42    522851 500 49344863345244392722361976152     .
60 .         .        -53      -42  52 2511.5002493448334524430027 2361976 152     .
61 .         .         . 53 -42  .  52 251 500 493448314534439 27 23 61916 12      .
62 .         .         .-53 -42  .52  511 500 49474834534443  27 23 61976. 12      .
63 .         .       -53-42      52  511 5004994748451 4432  27423 319716. 15      .
64 .        .53     -42        52  5118500494748453449432  27423 3197 16.  15     .
65 .        -42     -53    -352 . 5115040497648429  434   27423 3119 16 .  15     .
66 .       -53    -42  .     52  514150549-44483143142    27 2335119  16 .  215    .
67 .         .    46  42   53 52 514500495348  --4430. 8127323 51971 16  .  2 15   .
68 .         .         -521-531 504-4494648-45 ---4430 27 2356 19  1 16  .  2 15   .
69 .         .         .   5352515044944863-45-433-44075423-3119   1 16  . 2  15   .
70 .         .  -42    -51--53349.39348-476-45-431-4401238 119-3    116  . 2  15   .
71 .         .         4253251004934832 -47-45-43-44023411195  -3   116  . 2  15   .
72 .         .         . 53210493487343 -4745--4-443 -119086-5 .-3 1 16  . 2  15   .
73 .         .      532150490948-36-3733-4745-4-44  194108-6-5 . -31 16  .2   15   .
74 .         .  48    -4--53098  .-3-373-475--4411914110 -65   . -3  16  .2  15    .
75 .   53 52 51 50 49  48  4218  .-3-373-475-449-14110  -65    .-3   16  .2  15    .
76 .         .     531948 428-35 3-3-373-47-44091107 -65       -3   16   .2  15    .
77 -49    ---538      --42      -35 --37-47441029 -65.      1 -3    16   2   15    .
78 .        421035398  .      34-35 --37-474313095   .    1  -3.    16   2  15     .
79 .         .-11      34  1253  -3-3-37-47431309    .  1   -3 .    16   2  15     .
80 .         .       -53 42  24-343-3--3-47433029    1     -3  .   16    2  15     .
81 .         .         .-534244 3-3-3--3-4743309    1.     -3  .   16   2. 15      .
82 .         .  53   42.    2433 -34---3-4743309  1  .    -3   .   16   2. 15      .
83 .         .  33-42  -534      .-34--3-4743309 1   .    -3   .   16   2. 15      .
84 .         .       53. -42   24. ----3-47430291    .    -3   .  16    2.15       .
85 .         .   -42   .-53    -33 -3--3-47330291    .   -3    .  16   2 .15       .
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86 .         .      53 .   42   -33-3--3-473309 1    .   -3    .  16   2 .15       .
87 .         .     -53 . 42      -33-3-3-4730291     .   -3    .  16   2 .15       .
88 .         .     53  . 42 32   .-3-3---4730291     .   -3    . 16   2  15        .
89 .         .    -5342. -32     . -3----473029      .  -3     . 16   2  15        .
90 .         .         -5332  42 . -3----4730291     .  -3     . 16   2  15        .

RATES  1
   LOWER    RU         UPPER    NET        LOWER    RU         UPPER    NET        LOWER
   HE11S1 1.738E+07    HE12P1-4.784E+05    HE12S1 2.292E+07    HE12P1 5.726E+06    HE12P1 2.
   HE12P1 2.921E+07    HE13D1-2.164E+07    HE11S1 1.874E+06    HE13P1-1.273E+05    HE12S1 1.
   HE13S1 5.953E+06    HE13P1 1.149E+06    HE13D1 1.182E+04    HE13P1 4.169E+02    HE12P1 7.

POPS   1

   LEVEL    N          LEVEL    N          LEVEL    N          LEVEL    N          LEVEL
   HE11S1 8.495E-11    HE12S1 4.577E-13    HE12P1 8.251E-13    HE13S1 1.373E-13    HE13P1 2.
   HE14S1 9.392E-14    HE14P1 2.382E-13    HE14D1 4.083E-13    HE14F1 6.128E-13    HE15N1 1.
   HE17N1 3.527E-12    HE18N1 4.514E-12    HE12S3 2.275E-12    HE12P3 6.312E-12    HE13S3 6.
   HE13D3 2.565E-12    HE14S3 4.565E-13    HE14P3 1.296E-12    HE14D3 1.993E-12    HE14F3 2.
   HE16N3 1.137E-11    HE17N3 1.471E-11    HE18N3 1.853E-11    HE21   1.041E+02    HE22   1.
   HE24   1.534E-06    HE25   1.262E-06    HE26   1.256E-06    HE27   1.399E-06    HE28   1.
   HE210  2.305E-06    HE211  2.709E-06    HE212  3.190E-06    HE213  3.720E-06    HE214  4.
   H11    2.115E-05    H12    1.152E-06    H13    1.404E-06    H14    2.142E-06    H15    3.
   H17    6.030E-06    H18    7.743E-06    H19    9.689E-06    H110   1.186E-05    H21    4.


RYBIKI    1   6  1.67     0.100
1SURFACE FLUX

  IJ      FREQUENCY WAVELENGTH           HNU  M(TAU=1)    TAUMAX    ALBEDO

  1    1.0000000E+17        29.97925     2.533E-45 6.512E-02 1.861E+02     0.000
  2    8.0000000E+16        37.47406     2.838E-45 4.774E-02 2.314E+02     0.000
  3    6.0000000E+16        49.96542     5.713E-34 3.076E-02 3.605E+02     0.000
  4    5.0000000E+16        59.95850     3.129E-28 2.161E-02 5.219E+02     0.000
  5    4.0000000E+16        74.94812     1.327E-22 1.286E-02 8.741E+02     0.000
  6    3.4126250E+16        87.84806     2.125E-19 8.564E-03 1.303E+03     0.000


PROFILES


  1 RBB     1 HE      1 HE11S1  3 HE12P1 JS   78   86  HMAX 2.80931E-03  TAUL 7.656E-02  TAU

WAVE      583.843      583.877      583.912      583.929      583.947      583.964      583.
PROFILE  0.539160     0.554631     1.000000     0.891585     0.834374     0.891426     0.999

WAVE      584.051
PROFILE  0.539443


RATES


      RU          NET         RU          NET         RU          NET         RU          NE
    1 RBB       1 HE        2 RBB       1 HE        3 RBB       1 HE        4 RBB       1 HE
ID  1 HE11S1    3 HE12P1    2 HE12S1    3 HE12P1    3 HE12P1    4 HE13S1    3 HE12P1    6 HE1
 1  1.2943E+07 -5.4726E+05  1.3275E+07  4.6674E+06  2.5073E+06 -2.4046E+06  2.8121E+07 -3.05+05
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POPULATIONS

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       N           N/N*        N           N/N*        N           N/N*        N           N/
ID  1 HE        1 HE11S1    1 HE        2 HE12S1    1 HE        3 HE12P1    1 HE        4 HE1
 1  8.6783E-11  4.2527E-01  4.6823E-13  1.5960E+00  6.3727E-13  8.7651E-01  1.2453E-13  8.81-01
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1LOG CHANGES IN RATES

```
ID -8        -7        -6        -5        -4        -3        -2        -1         0
90 -89447--95987-73   896849099199311      .  -76    .         .         .         .
89 -35    -876498976594.  66899099931      .  -76    .         .         .         .
88 -35    -87510-746   -8-98979654  -6-73-9316-695   .-76      .         .         .
87 -35       .-87510 --77--98976594 -6-7-932-6-695   .-76      .         .         .
86 -35       .   -876  .3975-9897965462---930--697   .-76      .         .         .
85 -35       .       -875-375-9897654-6---930--6975  . -76     .         .         .
84 -354      .        -8769475-989765-62--9303--695  . -76     .         .         .
83 -35174    .       39.-8756755-98765----9383--697  . -76     .         .         .
82 -35   -17 . 39      47-8763755-98765---9380---695 .  -76    .         .         .
81 -3547-20  -17    -39. 5-8767505-98764--93806--695 .  -76    .         .         .
80 -35-255  -20   -17  47  -87675--98764---9386---69 .  -76    .         .         .
79 -3532-282515 -20   -17473-876757-9875----9386--696.   -76   .         .         .
78 -354-2-32-28-25513-208  -1-8775--9875--73-9386--69.   -76   .         .         .
77 .3531272421-32-2825158840 -94-98793 -73170 -938--69   -76   .         .         .
76 .33   353127242-329-28381-87666479876594.    -939-695 -76   .         .         .
75 .33292623 . 35317242-32-876251640275936398654 20--93695 -76  .         .         .
74 .      33292623  35387617-321-640512575036749864 3--9396-76 -1         .         .
73 .         . 332926 23  353018754074-259753967419864-9389-76 .  -1     .         .
72 .         .      3329 2623  356875  -3259753467429864-938-76-5    -1  .         .
71 .         .         . 33292623.640875 2259175457411985493876 19-54   -1.         .
70 .         .         .      339.26264087359 5754574759864939867 19-5 4  -1        .
69 .         .         .         .33292626487930754743129874932861798 5  4.-1       .
68 .         .         .       22.     339 2648735757433198649328671 3-14 .4-1      .
67 .         .         .        .        33292687675743-8985-93985911358.--1       .
66 .         .         .         .   -32   . 33292687754829864-938651913 8 41 -14  .
65 .         .         .         . 28      .   333916877530984--938651913.8 41  -14.
64 .         .         .         .         28   82329287554985-7-93865193-8 41    -14
63 .         .         .         .         .-2-82 3325968759866--93986-193 841     -14
62 .         .         .         .         .-70-829032926879874-7-93876-19 84-2    .-14
61 .         .         .         .         .82  -572731--687984-7-932876193841-2   .-14
60 .         .         .         .         .        829-3-789843---938-769384-1-2  .-14
59 .         .         .         .         .        3-822-329863-7-9318761984 -1-2 .-14
58 .         .         .         .         .       3-8226-998776-7-9328-769384 -1-2.-14
57 .         .         .      -59.         .       -94298962848757-93297-76384  --2.-14
56 .         .         .         .         .      33295-98-975-947--93921-7684   --2-14
55 .         .         .         .         .      332836284 -9897969439321-768    ---14
54 .         .         .         .         .     332984323 593-989796594321-765   -1-14
53 .         .         .         .         .-84  3329-882-8359--989796594271-76    ---14
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RYBIKI LINEARIZATION SOLVED BY BLOCK SOR

    1 ITERATIONS     6 SORS  OMEGA    1.670000     TOLERENCE  0.100


CORRECTIONS ITERATED   6 TIMES

MAXIMUM CHANGES IN DZ ARE
  1 62 -1.3E-01  2 54 -1.7E-01  3 81  1.5E-01  4 88  4.0E+00  5 62 -1.9E-01  6 82  5.8E-01  0
  9 56 -1.5E-01 10 77  1.9E+01 11 57 -1.9E-01 12 28  1.7E-01 13 82  3.9E-01 14 83  8.2E-02 11
 17 86  1.6E+00 18 64 -4.7E-01 19 60 -3.9E-02 20 56 -6.1E-01 21 62 -4.8E-02 22 68  8.1E+00 22
 25 64 -7.1E-01 26 27  4.5E-01 27 65 -5.7E-02 28 65 -5.5E+01 29 27  4.2E-01 30 27  4.0E-01 31
 33 27  3.8E-01 34 27  3.7E-01 35 28  1.6E-01 36 90 -1.2E+00 37 73 -1.4E-01 38 76  2.9E+01 30
 41 89 -2.4E-01 42 76  1.1E+01 43 77  8.7E-01 44 71  2.3E+00 45 73 -1.3E-01 46 76  8.3E+00 40
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 49 89 -2.3E-01 50 62  8.6E+00 51 77  9.1E-01 52 71  2.6E+00 53 73 -1.2E-01 54 62  1.1E+01 50
 57 89 -2.2E-01 58 62  1.5E+01 59 57  3.4E+02 60 71  2.7E+00 61 73 -1.1E-01 62 62  2.3E+01 60
 65 54 -4.2E-01 66 85  1.2E-01 67 90  1.8E-01 68 90  1.2E-01 69 89  2.0E-01 70 62  2.8E+01 70
 73 61  4.5E+00 74 88  3.0E+01 75 85 -4.4E-01 76 48 -9.8E-02 77 89  1.1E-01 78 89  1.4E-01 71
 81 86 -2.4E-01 82 61 -7.8E+00 83 77  6.9E+00 84 53  9.8E+00 85 74 -5.9E-01 86 74 -6.2E-01 80
 89 45  3.0E-01 90 47  2.6E-01 91 45  1.9E-01 92 82 -1.6E-01 93 82 -1.7E-01 94 57 -3.2E+00 90
 97 57 -1.8E+00 98 57 -1.9E+00

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